

EduGenie:Google Gemini Powered Learning Assistant

If you mean “**EduGenie: Google Gemini Powered Learning Assistant**” as a project/reference, I found several similarly named EduGenie projects. The one matching your wording most closely is an AI-powered learning web application described as using **Google Gemini 1.5 Pro + FastAPI**. Its reported features include AI question answering, simple topic explanations, text summarization, automatic quiz generation, and personalized learning-path recommendations.

Detailed reference

Project title: EduGenie – Google Gemini Powered Learning Assistant

Purpose:

EduGenie is designed as a personalized AI learning assistant that helps students understand academic content, revise efficiently, test their knowledge, and follow a personalized learning path.

Core features

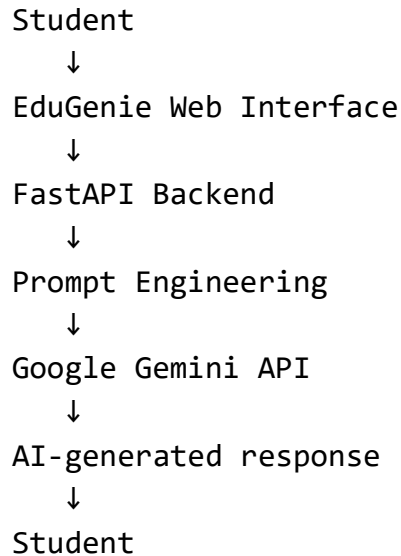
1. **AI Question & Answer** – Students can ask academic questions and receive AI-generated answers.
2. **Topic Explanation** – Explains difficult concepts in simpler language.
3. **Text Summarization** – Converts lengthy educational material into concise summaries.
4. **Quiz Generation** – Automatically creates questions for self-assessment.
5. **Personalized Learning Path** – Recommends learning steps based on the student's needs.
6. **Responsive Web Interface** – Designed to work across different screen sizes.

Technology stack reported for this version

- **AI:** Google Gemini 1.5 Pro
- **Backend:** FastAPI
- **Programming:** Python
- **Frontend:** HTML, CSS, JavaScript / Jinja2
- **AI integration:** Google Gemini API
- **Architecture:** REST API-based web application

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How the system works



For example, a student could enter:

“Explain photosynthesis in simple terms.”

EduGenie sends the request to the Gemini-powered backend, processes the response, and presents an understandable explanation. Similarly, study material can be summarized or transformed into quiz questions.

Possible project modules

Module	Function
User Interface	Student interaction with EduGenie
AI Q&A	Answers academic questions
Explanation Engine	Simplifies difficult concepts
Summarizer	Creates concise study notes
Quiz Generator	Generates practice questions
Learning Path	Suggests personalized study progression
Gemini Service	Communicates with Google Gemini
FastAPI Backend	Handles API requests and application logic
Database/Auth	User accounts and learning information
// =====	

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```
// Elements
// =====

const task =
  document.getElementById("task");

const level =
  document.getElementById("level");

const levelLabel =
  document.getElementById("level-
label");

const input =

document.getElementById("inputText");

const inputLabel =
  document.getElementById("input-
label");

const submitBtn =

document.getElementById("submitBtn");

const status =
  document.getElementById("status");

const resultCard =

document.getElementById("resultCard");

const result =
  document.getElementById("result");

// =====
// Task Information
```

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```
// =====
```

```
const taskInfo = {

  qa: {

    label: "Your Question",

    placeholder:
      "Example: Which is the largest
ocean?",

    button:
      "Ask EduGenie"
  },

  explain: {

    label: "Topic to Explain",

    placeholder:
      "Example: Explain photosynthesis
for a beginner.",

    button:
      "Explain Topic"
  },

  quiz: {

    label:
      "Topic or Educational Passage",

    placeholder:
      "Paste a topic or educational
passage to generate 3 MCQs.",
```

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```
    button:
      "Generate Quiz"
  },
```

```
  summarize: {
```

```
    label:
      "Text to Summarize",
```

```
    placeholder:
      "Paste a long educational passage
here.",
```

```
    button:
      "Summarize"
  },
```

```
  learn: {
```

```
    label:
      "Topic to Learn",
```

```
    placeholder:
      "Example: SQL",
```

```
    button:
      "Build Learning Path"
  }
```

```
};
```

```
// =====
```

```
// Update Form
```

```
// =====
```

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```
function updateForm() {

    const info =
        taskInfo[task.value];

    inputLabel.textContent =
        info.label;

    input.placeholder =
        info.placeholder;

    submitBtn.textContent =
        info.button;

    const isLearning =
        task.value === "learn";

    level.classList.toggle(
        "hidden",
        !isLearning
    );

    levelLabel.classList.toggle(
        "hidden",
        !isLearning
    );
}

task.addEventListener(
    "change",
```

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```
        updateForm
    );

    updateForm();

    // =====
    // Escape HTML
    // =====

    function escapeHtml(value) {

        return String(value)

            .replaceAll(
                "&",
                "&amp;"
            )

            .replaceAll(
                "<",
                "&lt;"
            )

            .replaceAll(
                ">",
                "&gt;"
            )

            .replaceAll(
                "'",
                "&quot;"
            )

            .replaceAll(
                '"',
                "&#039;"
            )
    }
}
```

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```
    );
  }

// =====
// Render Quiz
// =====

function renderQuiz(
  questions
){

  result.innerHTML =
    questions
    .map(
      (question, index) => `

        <div class="quiz-question">

          <strong>
            ${index + 1}.
            ${escapeHtml(
              question.question
            )}
          </strong>

          ${question.options
            .map(
              option => `

                <button
                  class="quiz-option"
                  data-
answer="${escapeHtml(
question.correct_answer
                )}"
```

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```
        data-
option="${escapeHtml(
        option
    )}"
    >
        ${escapeHtml(
            option
        )}
    </button>

    ,

)
.join("")}

<p class="quiz-feedback"></p>

</div>

,

)
.join("");

document
    .querySelectorAll(
        ".quiz-option"
    )
    .forEach(
        button => {

            button.addEventListener(
                "click",
                () => {

                    const parent =
                        button.closest(
                            ".quiz-question"
                        );
```

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```
const isCorrect =
  button.dataset.answer ===
  button.dataset.option;

parent
  .querySelectorAll(
    ".quiz-option"
  )
  .forEach(
    option => {
      option.disabled =
        true;
    }
  );

button.classList.add(
  isCorrect
    ? "correct"
    : "wrong"
);

const feedback =
  parent.querySelector(
    ".quiz-feedback"
  );

feedback.textContent =
  isCorrect
    ? "Correct!"
    : `Correct answer:
    ${button.dataset.answer}`;
}
```

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```
        );

    }

    );
}

// =====
// Submit
// =====

async function submit() {

    const text =
        input.value.trim();

    if (!text) {

        status.textContent =
            "Please enter some text first.";

        return;
    }

    submitBtn.disabled =
        true;

    status.textContent =
        "EduGenie is thinking...";

    resultCard.classList.add(
        "hidden"
    );
}
```

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```
let url;
```

```
let body;
```

```
// -----
```

```
// Q&A
```

```
// -----
```

```
if (  
    task.value === "qa"  
) {
```

```
    url = "/qa";
```

```
    body = {  
        question: text  
    };
```

```
}
```

```
// -----
```

```
// Explanation
```

```
// -----
```

```
else if (  
    task.value === "explain"  
) {
```

```
    url = "/explain";
```

```
    body = {  
        text: text  
    };
```

```
}
```

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```
// -----  
// Quiz  
// -----
```

```
else if (  
    task.value === "quiz"  
) {
```

```
    url = "/quiz";
```

```
    body = {  
        text: text  
    };
```

```
}
```

```
// -----  
// Summary  
// -----
```

```
else if (  
    task.value === "summarize"  
) {
```

```
    url = "/summarize";
```

```
    body = {  
        text: text  
    };
```

```
}
```

```
// -----  
// Learning Path
```

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```
// -----

else {

    url =
        "/learn/recommendations";

    body = {

        topic: text,

        level:
            level.value
    };
}

try {

    const response =
        await fetch(
            url,
            {

                method: "POST",

                headers: {
                    "Content-Type":
                        "application/json"
                },

                body:
                    JSON.stringify(body)
            }
        );

    const data =
```

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```
        await response.json();

        if (!response.ok) {

            throw new Error(
                data.detail ||
                "Request failed."
            );
        }

        // -----
        // Quiz result
        // -----

        if (
            task.value === "quiz"
        ){

            renderQuiz(
                data.quiz
            );

        }

        // -----
        // Text result
        // -----

        else {

            const value =
                data.answer ||
                data.explanation ||
                data.summary ||
                data.recommendations;
```

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```
        result.innerHTML =
        `
        <div class="result">
            ${escapeHtml(value)}
        </div>
        `;
    }

    resultCard.classList.remove(
        "hidden"
    );

    status.textContent =
        "Done.";

}

catch (error) {

    result.innerHTML =
    `
    <div class="result">
        ${escapeHtml(
            error.message
        )}
    </div>
    `;

    resultCard.classList.remove(
        "hidden"
    );
}
```

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```
        status.textContent =  
            "Something went wrong.";  
    }
```

```
finally{
```

```
    submitBtn.disabled =  
        false;  
}  
}
```

```
// =====  
// Button Event  
// =====
```

```
submitBtn.addEventListener(  
    "click",  
    submit  
);
```

```
// =====  
// Ctrl + Enter  
// =====
```

```
input.addEventListener(  
    "keydown",  
    event => {  
  
        if (  
            event.ctrlKey &&  
            event.key === "Enter"  
        ){  
  
            submit();
```

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```
    }  
  }  
);  
ifhu static/style.css  
<!DOCTYPE html>  
<html lang="en">  
  
<head>  
  
  <meta charset="UTF-8">  
  
  <meta  
    name="viewport"  
    content="width=device-width, initial-  
scale=1.0"  
  >  
  
  <title>  
    EduGenie - AI Learning Assistant  
  </title>  
  
  <link  
    rel="stylesheet"  
    href="{{ url_for('static',  
path='style.css') }}"  
  >  
  
</head>  
  
<body>  
  
  <main class="container">  
  
    <!-- Header -->  
  
    <header class="hero">
```

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```
<div class="badge">  
  AI LEARNING ASSISTANT  
</div>
```

```
<h1>  
  EduGenie  
</h1>
```

```
<p>  
  Learn smarter with AI.  
  Ask questions, understand  
concepts,  
  generate quizzes, summarize  
content,  
  and build learning paths.  
</p>
```

```
</header>
```

```
<!-- Input Card -->
```

```
<section class="card">
```

```
<label for="task">  
  Choose a Task  
</label>
```

```
<select id="task">
```

```
<option value="qa">  
  Ask a Question  
</option>
```

```
<option value="explain">  
  Explain a Topic  
</option>
```

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```
<option value="quiz">  
  Generate Quiz  
</option>
```

```
<option value="summarize">  
  Summarize Text  
</option>
```

```
<option value="learn">  
  Learning Path  
</option>
```

```
</select>
```

```
<!-- Learning Level -->
```

```
<label  
  for="level"  
  id="level-label"  
  class="hidden"  
>  
  Learning Level  
</label>
```

```
<select  
  id="level"  
  class="hidden"  
>
```

```
<option value="beginner">  
  Beginner  
</option>
```

```
<option value="intermediate">  
  Intermediate  
</option>
```

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```
<option value="advanced">
  Advanced
</option>
```

```
</select>
```

```
<!-- Input -->
```

```
<label
  for="inputText"
  id="input-label"
>
  Your Question
</label>
```

```
<textarea
  id="inputText"
  rows="8"
  placeholder="Example: Which is
the largest ocean?"
></textarea>
```

```
<!-- Submit -->
```

```
<button
  id="submitBtn"
>
  Ask EduGenie
</button>
```

```
<p
  id="status"
  class="status"
  aria-live="polite"
></p>
```

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</section>

<!-- Result -->

<section
 id="resultCard"
 class="card result-card hidden"
>

<h2>
 Result
</h2>

<div id="result"></div>

</section>

<!-- Footer -->

<footer>

EduGenie
•
FastAPI + Google Gemini

</footer>

</main>

<script
 src="{{ url_for('static',
path='app.js') }}"
></script>

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</body>

</html>

ifhu template/index.html

EduGenie setup definitions

1. Install Python

- a. Python is the programming language/runtime required to run EduGenie's backend, AI logic, scripts, and supporting services.
- b. Typical verification:

```
python --version
```

2. Install PowerShell

- a. PowerShell is a Windows command-line shell used to run installation and management commands.
- b. It can be used to create virtual environments, install packages, configure variables, and start EduGenie.
- c. Verify with:

```
$PSVersionTable.PSVersion
```

3. Install Bash

- a. Bash is a Unix/Linux command shell commonly used by development scripts.
- b. On Windows, **Git Bash** or **WSL** can provide Bash.
- c. Verify with:

```
bash --version
```

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4. Install all dependencies

- a. This means installing every software library/package required by EduGenie.
- b. For a Python project, this is commonly done from `requirements.txt`:

```
pip install -r requirements.txt
```

- c. If the project uses a `pyproject.toml`, installation may instead use:

```
pip install -e .
```

5. Start EduGenie

- a. This means launching the EduGenie application after Python, the required shell tools, and dependencies are configured.
- b. The exact command depends on EduGenie's project structure.
Common examples are:

```
python app.py
```

or

```
python main.py
```

or, for a web application:

```
uvicorn main:app --reload
```

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Overall definition

“Install Python → Install PowerShell/Bash → Install all dependencies → Start EduGenie” describes the **development environment setup and application startup workflow** for EduGenie.

If you have the **EduGenie project folder/repository**, I can also give you the exact Windows commands to install everything and start it, rather than using generic commands.

There are also other EduGenie implementations using **RAG, PDF processing, vector search, Streamlit, and Firebase**, so if you're preparing a **college project report**, it's important not to mix their architectures. One such implementation specifically converts PDFs into notes, flashcards, MCQs, question banks, mock tests, and revision sheets using RAG. Another 2026 research implementation uses Gemini, Streamlit, Firebase Firestore, adaptive quizzes, and Bloom's Revised Taxonomy.

If by **“referens detailed”** you mean you want a **detailed project reference/documentation** for *EduGenie – Google Gemini Powered Learning Assistant*, I can prepare the full **abstract, introduction, objectives, existing system, proposed system, architecture, modules, methodology, technology stack, database design, API flow, advantages, limitations, future scope, and references** in project-report format.



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